

- k) UPS/UPSS/CWS/CDS unemployment rates (can be easily estimated from the tables on distribution of 1000 persons by various characteristics, as indicated in the main paragraph) for all groups together and for different groups like gender, age group, educational level, MPCE classes, household type, household land cultivated class and principal household economic activity, etc.;
- l) Average daily salary/wage earnings of regular wage/salaried employees aged 15-59 by sex, sector of work and education; a similar distribution by occupational groups and education; and
- m) Average daily wage/earnings per day received by casual labour by sex, age group, type of operation, and sub-round¹⁵.

Data on employment are thus available from several sources. Estimates of employment derived from different sources for the same sector or sub-sector will differ. While this may be frustrating to a lay person, it opens up an exciting and challenging opportunity to an analyst to unravel the factors responsible for such variations and to find ways and means of tackling such factors. Such factors could be differences in the concepts, definitions and mode of collection of data (sample survey, census or administrative/statutory reporting systems) used by the different sources. There may be other factors causing the divergence in estimates. Some steps could be taken to remove at least a few of these factors. Data collecting agencies should adopt the same concepts and definitions for enquiries on the same or similar subjects, as far as possible. The use of trained personnel in data collection does have a favourable impact on the quality of data and should receive consideration. Estimates based on sample surveys should be accompanied by estimates of the standard error to which the estimates from the survey are subject¹⁶.

24.2.2 Quality/Adequacy of Employment

NSSO survey data (items above) can be used to analyse the quality and adequacy of employment. Besides underemployment among the employed, other aspects of quality can also be looked at. One is the proportion of the employed by employment status, especially the proportion of the work force in 'casual employment'. The second is a comparison of the average daily earnings of male and female, regular and casual workers in different sectors and operations with the prevailing statutory minimum wage and the poverty line. The third is a look at the pattern of employment status of the workers belonging to poor households. These will throw light on the quality of employment enjoyed by the workforce, quality in terms of the intermittent nature of work, tenure of employment, employment security and low wages and above all, the prevailing gender differences in these aspects of quality. Gender differences in access to employment in different occupations and sectors of economic activity would be clear from a perusal of the tables listed in the preceding section. Even aspects of the question of the invisibility of the female worker, leading to underestimation of the female workforce can be examined¹⁷.

¹⁵ One year long round of the Survey has four sub-rounds to take note of the seasonal effect on the variables under investigation.

¹⁶ Reports on NSSO surveys do provide information on the concepts, definitions, coverage, sampling design and estimation procedures adopted and the standard errors of estimates presented in the reports.

¹⁷ NSSO Report No. 554 referred to in an earlier footnote. See also "Gender and Employment in India", T.S. Papola & Alakh N. Sharma (Ed), published by Indian Society of Labour Economics and Institute of Economics Growth in association with Vikas Publishing House Ltd, New Delhi. (1999).

The Labour Bureau, Shimla under the Ministry of Labour also publishes¹⁸ data on wage levels in the organized sector and on the welfare of labour. Data on total earnings of factory workers are collected through the statutory returns under the Payment of Wages Act, 1936 from establishments defined as factories under the Factories Act, 1948. The Act was applicable to employees with earnings¹⁹ up to Rs. 200. This limit has been increased from time to time to 1600 in 1982. These returns are, however, not received from all factories. The Bureau also conducts Occupational Wage Surveys in selected industries at regular intervals (called first round, second round and so on) and publishes reports²⁰ on these surveys) that provide the distribution of workers by levels of earnings and by occupations for each industry. These reports facilitate analysis of variation in levels of earnings within occupations, across gender within occupations, across regions and industries and to judge the adequacy of such employment opportunities in the light of statutory minimum wages and the poverty line. We have seen in the Unit on Agricultural and Industrial Data that the ASI gives information on the total wages, total emoluments and employee compensation along with mandays of workers and mandays of employees in different industries in the factory sector. These thus help in deriving the average wage per worker-manday and the average emoluments per employee-manday and the average employee compensation per employee-manday.

As for the unregistered sector and the unorganized part of other sectors, the **DE, NDE and OAE Surveys²¹ and other Establishment Surveys of the CSO** in different sectors (till the early 1990s) and the **unorganized sector surveys of the NSSO²²** provide data on average annual earnings for men, women and children in such establishments. **Regular reports like “Wage Rates in Rural India”** (the latest published relates to 2004-05) and the **Report on the Working of the Minimum Wages Act, 1948** (the latest is for 2003), of **Labour Bureau** enable an analysis of rural/unorganized sector wage levels *vis-à-vis* statutory minimum wages and poverty line.

The prevalence of child labour, which depicts an exploitative dimension of the economic system, can be seen from the Census and NSSO tables on the distribution of the work force by age groups and activity, the former down to the district level and the latter up to the State level (technically down to NSS regions due to availability of the unit record data on floppies/CDs). Such data may not fully reflect the ground level realities because surveys may not be able to extract information on employment of children, for various reasons. Further, such data are also dated. These can help only in drawing attention to the areas where child labour is prevalent so that the authorities concerned can initiate further action. It is usually the Non-Government Organisations (NGOs) active in the field of the rights of children that succeed in locating establishments employing children and press the Government to take further action.

¹⁸ **Yearbooks of Labour Statistics, (various years), Ministry of Labour, Shimla.**

¹⁹ Earnings include basic wages, dearness allowance, money value of concessions, annual or prepaid bonus, and arrears.

²⁰ **Reports on Occupational Wage Surveys** (Successive Rounds – the latest publications relate to the sixth round), Ministry of Labour, Labour Bureau, Shimla.

²¹ Directory Establishments, Non Directory Establishments and Own Account Enterprises. The first two employ at least one hired worker and a total of 6 or more persons and 1 to 5 persons respectively. In Own Account Enterprises, no hired worker works on a regular basis. These surveys are now being conducted by the NSSO. The of **NSS Report 549, for the year 2010-11 is the latest report on unorganized non agricultural enterprises (excluding construction), which provide data on DE, NDE and OAE.**

²² Besides the reports listed in Sub-section 21.3.4 of unit 21.

24.2.3 Labour Welfare

The Labour Bureau publishes²³ data on several aspects of labour welfare – data on industrial injuries, injuries in mines, compensation to workers for injuries and death, industrial disputes, health insurance, provident fund and trade unions of employers and workers. Statistics on industrial injuries are collected through statutory returns under the Factories Act, 1948 that provides that industrial accidents due to which the affected persons are prevented from attending to work for at least 48 hours should be reported to the inspector of factories. These depend on returns, and the sizeable non-response in the submission of these mars the quality of this source of data on incidence of fatal and non-fatal injuries in factories in mines under the Mines Act, 1952, as amended in 1983. Initially, data on the number of serious accidents and the accident rate were classified as ‘fatal’ and ‘serious’. Subsequently, from 1984, accidents and the accident rate (accidents per 100 employees) were classified as ‘fatal’ ‘spot serious’²⁴ and ‘reportable serious’²⁵. Complete statistics are not available for the same reasons mentioned earlier. Statistics on ‘compensated injuries’ and the amount of compensation paid, both classified as (resulting in) ‘death’ and (leading to) ‘disability’ – ‘permanent’ and ‘temporary’, are collected under the Workmen’s Compensation (WC) Act, 1923 on the basis of annual returns received from the State Governments, Posts & Telegraph Departments and the Railway Board for different Zonal Railways under the Act. Compensation is payable to workers employed in ‘scheduled employments’²⁶ for injuries due to accidents resulting in death or disablement for more than three days, provided that it is not caused through the fault of the worker himself. The number of injuries reported in the returns does not reflect the total number of injuries that occur, as all the injuries are not compensated for. Further, many of the establishments covered by the Act fail to submit returns and, therefore, the information received by the State Governments about the compensated injuries and the amount of compensation paid is incomplete. Compensation for injuries in establishments covered by the Employees State Insurance (ESI) Act, 1948 are paid under the ESI Act and not under WC Act. These limitations affect the trends and comparability over time of compensated accidents.

Statistics on the number of (a) industrial disputes²⁷, (b) workers involved in the disputes, (c) mandays lost and (d) causes of disputes – (i) wages and allowances, (ii) bonus, (iii) personnel matters, (iv) retrenchment, (v) leave and hours of work/shift working, (vi) indiscipline and violence, (vii) others, and (viii) cause not known. Completeness of these data would depend on the extent to which outside authorities are involved in the resolution of the disputes. How are the workers and the employers, the two pillars of industrial progress, organized to fight for their rights? The Trade Unions Act, 1926 regulates the formation and functioning of such organizations. The annual returns under the

²³ Yearbooks of Labour Statistics referred to in an earlier footnote.

²⁴ This is one where one or more persons have received serious bodily injury. This means any injury which involves, or in all probability will involve, (a) the permanent loss of any part or section of the body or use of any part or section of the body or (b) the permanent loss of, or injury to, sight or hearing or (c) any permanent incapacity or (d) the fracture of any bone or one or more joints or bones of any phalanges of hand or foot.

²⁵ This is one where one or more persons have received “reportable injury”. This means any injury other than a serious bodily injury, which involves, or in all probability will involve, the enforced absence of the injured person from work for a period of 72 hours or more.

²⁶ Employments specified in the Schedule appended to the Act under reference.

²⁷ Disputes resulting in work-stoppages involving 10 or more workers.

Act received from the State Governments relate only to those organizations that have been registered with the State Governments under the Act. And it is not obligatory on these organizations to secure registration under the Act. The response rate even among the registered ones is less than 50 per cent. The data provided in the **Yearbooks of Labour Statistics** relates to the number of (i) workers' organizations on the register (ii) those submitting returns, (iii) percentage response, (iv) membership at the end of the year, (v) income including balance carried over from the previous year, (vi) expenditure, and (vii) balance of funds at the close of the year and similar data for employers' organizations. Information on the membership of the employers' and workers' organizations by NIC 1987 is available in the Yearbooks. Statistics relating to the working of various welfare funds like the Coal, Mica etc., Mines Welfare Funds and the ESI Corporation and the Employees Provident Fund Organisation (EPFO) – number of beneficiaries/members, etc., - are also provided by Yearbooks and the **annual reports of the ESIC and EPFO**²⁸.

As for the unorganized sector, the **Labour Bureau's** reports on their ongoing programme of surveys on (i) the working and living conditions of Scheduled Caste/Tribe workers, (ii) living conditions of unorganized workers, and (iii) contract labour provide information on the qualitative aspects of employment in terms of variables like wage levels, working and living conditions, work place safety and amenities at the work place of these classes of unorganized workers in rural and urban areas.

Check Your Progress 1

- 1) Name the sources that provide data on employment.

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- 2) Explain the kind of data on employment and unemployment available from the population census 2001.

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- 3) State the different measures of employment/unemployment/labour force used by NSSO in different quinquennial surveys.

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²⁸ See also the Section on Health in this Unit.

- 4) Identify the different aspects of labour welfare on which data are compiled by the Labour Bureau.

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24.3 EDUCATION

Education is an important instrument for empowering people, especially women. Education nurtures and develops their innate talents and capabilities and enables them to contribute effectively to the development process and reap its benefits. It is also an effective instrument for reducing social and economic inequalities. We have built up over the years a vast educational system in an attempt to provide education for all, to ensure that the skill and expertise needs of a developing economy are met satisfactorily and at the same time, to monitor the functioning of the educational system as an effective instrument for tackling inequalities. We would, therefore, like to look at data on different aspects of the educational and training system such as its size and structure, the physical inputs available to it for its effective functioning, its geographical spread, the type of skills and expertise it seek to generate, the access of different sections of society and areas of the country to it and the progress made towards the goals like 'total literacy', 'universalisation of secondary education', 'education for all' and 'removal of inequalities'.

The United Nations, in the year 2000, has put achieving of universal primary education as one of the eight Millennium Development Goals, to be achieved by 2015. Education plays an important role in shaping the post 2015 development agenda as well. The Government of India has enacted the Right to Education Act in August 2009. Therefore, the importance of research on the progress of the nation towards achieving these goals and its effects on the other social and economic goals is increasing by day.

The Department of School Education and the Department of Higher Education of the Ministry of Human Resources Development (MHRD), the National University of Educational Planning and Administration (NUEPA), National Council for Educational Research and Training (NCERT) and the University Grants Commission (UGC) collect and publish educational statistics, conduct research studies and surveys in the area of education. The International Standard Classification of Education (ISCED) is used worldwide to compile and compare cross-country statistics of education. In 2014, for the first time, the MHRD has developed the Indian Standard Classification of Education (InSCED), as a part of collection, dissemination and presentation of statistical data on education. Educational activities have been first classified into 16 broad levels from A through O and X. Levels A to E pertain to School Education, levels F to L pertain to Higher Education, levels M, N and O pertain to Certificate Courses, In-service training and Adult Education. The detailed structure can be accessed at http://mhrd.gov.in/sites/upload_files/mhrd/files/statistics/InSCED2014_0.pdf.

The annual publication “**Educational Statistics at a Glance**” provide a host of information on school education, like number of educational institutions, level and Stage-wise enrollment in school and higher education, teacher-student ratio, drop-out rates, examination results, public expenditure on education, etc. The MHRD also publishes a compendium on Universities in India, providing University-wise data on student enrollment, number of teachers, staff quarters, hostels, etc. The Annual Report of the MHRD also provides information on school and higher education in India. All these reports can be accessed free of charge at the website www.mhrd.gov.in in the sections documents and reports and the section statistics.

The NUEPA has developed a **District Information System of Education (DISE, can be accessed at www.dise.in)** to record data from the schools and junior colleges in all districts of India. Analytical reports based on this data, like school and facility related indicators, enrollment based indicators, teacher related indicators and EDI and analytical tables can be accessed at <http://www.dise.in/AR.htm>. These reports are available from the period 200-02 onwards, although the time series varies among topics. Provisional results of the “**Report of the eighth AIES**”, (<http://www.aises.nic.in/surveyoutputs>) published by the NCERT, for the year 2009, in spite of being a bit dated, provides important data on school education statistics. While the data in the publication of MHRD are based on returns from the State Governments and Union Territory Administrations, the last one is based on a field survey. The University Grants Commission and MHRD provide data on university level courses in professional and technical courses. The MHRD has set up a **National Technical Manpower Information System (NTMIS)** – (see http://www.iamrindia.gov.in/_aboutNTMIS.htm) with lead centre at the Institute of Applied Manpower Research (IAMR), New Delhi and 21 nodal centres located at different States. It provides data on technical manpower – intake in and output from institutions and the utilization patterns of such output in detail through Tracer Studies²⁹ and other studies. These are disseminated through reports released from time to time by IAMR. **The Employment-Unemployment surveys of the NSSO and Education surveys conducted in the 52nd (1995-96), 64th (2007-08) and 71st (2014) Round of NSSO and the social and cultural tables of the population Census (ORGI)** also provide useful data on literacy and educational composition of the population or stocks of educated manpower of different levels of education and their utilization patterns – snapshots, at specific points of time, of the impact of the efforts in the area of education.

The kind of data available in these publications and sources are indicated below:

24.3.1 Educational Infrastructure

- a) Levels of literacy and progress of efforts under literacy campaigns; (MHRD)
- b) The number of institutions – colleges (including universities, deemed universities and institutions of national importance) and all types of general, technical and professional schools; (MHRD & UGC)

²⁹ Tracer Studies are, as the name suggests, trace or follow up specific cohorts or batches of students passing out of institutions to find out their present activity status. This helps in evaluating the trends over time in (a) the capacity of the economy to absorb the specific category of manpower in employment, (b) the waiting time for employment for the category of manpower, and (c) market value, in terms of salary levels in first employment, of the manpower category. These surveys are also called Cohort Studies.

- c) The number of universities, deemed universities and institutions of national importance, the distribution of number of colleges by categories like those teaching arts, science & commerce, oriental learning and different categories of professional subjects like law, agriculture, engineering & technology and medicine and the distribution of schools in the area of general education (and stages like pre-primary and primary etc.,) and different categories such as those dealing with vocational, professional, special education etc. (MHRD & UGC);
- d) Teacher (male and female and those belonging to SC/SC communities), the proportion trained among the, terms and conditions of their employment and attrition of the stock of teachers by cause. (AIES);
- e) Patterns of management of colleges including professional colleges and deemed universities, recognition by appropriate bodies, availability of teachers and facilities like laboratory and equipment etc. (UGC);
- f) Patterns of management (Government, local bodies, private management, religious and linguistic minority trusts/organizations and so on), medium of instruction, type of school buildings (*pucca, kutcha*, thatched hut etc.), crowding (number of rooms in the building and the number used for instructional purposes), availability and adequacy of facilities like drinking water, laboratories and urinals (and whether these are available separately for girls). (Sixth AIES);
- g) Teachers in professional and technical institutions and the number of technical teacher training institutions. (MHRD & UGC);
- h) The number of Industrial Training Institutions (ITIs) and Advanced Training Institutes, and the training capacity for apprenticeship training in industry under the Apprenticeship Act, 1961 for those passing out of ITIs and the vocational stream of schools. (MoLE and MHRD);
- i) The number of institutions for training instructors for it is. (MoLE)³⁰;
- j) The number of Vocational Rehabilitation Centres (VRCs) for the Physically Handicapped³¹ set up by the DGE&T all over the country for giving adjustment training and placement in suitable employment through the Special Employment Exchanges for this group of people. (DGE&T³²);
- k) Expenditure on education by programmes like the *Sarva Siksha Abhiyan*, the Total Literacy Campaign and Adult Education; direct and indirect expenditure on recognized institutions of education. (MHRD).

24.3.2 Infrastructure Utilisation and Access to Educational Opportunities

- a) Literacy rates – overall and age-specific, among SC/ST and adults; (Census – Social & Cultural Tables; NSSO & MHRD);
- b) The number of students and the number of female students in the institutions specified in item ‘b’ above;

³⁰ Annual Reports of the Ministry of Labour, the part relating to DGE&T.

³¹ The phrase “physically challenged” instead of the phrase “physically handicapped” is being brought into use nowadays in public discussions on the subject.

³² See the preceding footnote.

- c) The number of students and the number of girl students by courses and stages of education in recognized institutions – from the nursery class to the high/higher secondary level and in schools for vocational & professional education and special education and similar data for rural areas. (MHRD);
- d) The number of male and female workers and marginal workers in the age group 5-9 and 10-14 – working children; (Census and NSSO); a comparison of this with the population in these age groups and data on school enrolment would lead to an assessment of the number of children who are neither in school nor in the workforce;
- e) Enrolment and output in institutions specified in items 'h', 'I' and 'j' above;
- f) Enrolment (and enrolment of female students) in university level general education courses by stages (degree, post graduate degree, diploma/certificate, research) and university level professional and technical education courses by faculty. (MHRD & UGC);
- g) Similar information of the kind mentioned above for Scheduled Castes and Tribes (SC/ST) and females belonging to such sections of society. (MHRD & UGC);
- h) Teacher-Pupil ratios at different levels of education. (MHRD);
- i) Intake and output of graduates and post graduates in different disciplines and faculties. (MHRD & UGC);
- j) Coverage of population in appropriate age groups by different stages of education. (MHRD);
- k) Dropout rates at different stages of education. (MHRD);
- l) Distribution of population attending educational institution by age, sex and type of educational institution for general and SC/ST population;
- m) Access of the general and SC/ST population to (or availability of) facilities for different levels of education in rural habitations and urban settlements belonging to different population slabs, in terms of distance of the habitation from the facility; similar information regarding non-formal education (NFE) centers. (Sixth AIES);
- n) Availability of special institutions suited to different types of disability of children in different villages and towns and the enrolment of disabled children in such institutions. (Sixth AIES);
- o) Utilization patterns of different categories of professional and technical manpower in general and trends in waiting time for employment, utilization patterns and bargaining power for employees' compensation (salary etc.,) through tracer studies. (IAMR);
- p) The disabled among main workers, marginal workers and non-workers by type of disability, age and sex. (Census);
- q) Stocks of educated manpower and also selected categories of technical manpower for the general population, SC/ST and by religious communities;

- r) The distribution of population by different levels of educational attainment (including illiterates, literates without any educational attainment) for population groups like SC/ST/Backward Classes (BC)/general households and those belonging to different religions, household types like (a) those with different sizes of land holdings; (b) households self-employed in agriculture, households self-employed outside agriculture, agricultural labour households, other labour households and households depending largely on regular wage/salaried employment outside agriculture; and (c) households belonging to different monthly per capita consumption expenditure (MPCE) classes. (NSSO and Census)³³.

One should not forget to mention in this context the **National Human Development Reports (NHDR)** prepared by the Planning Commission (now rechristened as NitiAyog) and the HDRs prepared by different State Governments as very useful sources for the manner in which available data on education, health, and other areas have been utilized to compute human development indicators and also as important sources for basic data on these sectors.

Check Your Progress 2

- 1) Name the organizations/institutions that collect and publish the data on different aspects of education.

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- 2) What do you understand by the term 'Tracer Studies'? How are these useful?

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- 3) Prepare a chart stating the kind of educational data available in the various documents published by MHRD and UGC.

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³³ See also the sub-section (b) on education in the Section on 'Level of Inequality on Non-income Aspects of Life in Unit 4 on Poverty & Inequality: Policy Implications in Block 1 of MEC – 005 on Indian Economic Policy in your first year course.